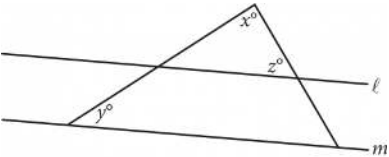


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure above, lines ℓ and m are parallel, $y = 20$, and $z = 60$. What is the value of x ?

Answer

- A. 120
- B. 100
- C. 90
- D. 80

Correct Answer: B

Rationale

Choice B is correct. Let the measure of the third angle in the smaller triangle be a° . Since lines ℓ and m are parallel and cut by transversals, it follows that the corresponding angles formed are congruent. So $a^\circ = y^\circ = 20^\circ$. The sum of the measures of the interior angles of a triangle is 180° , which for the interior angles in the smaller triangle yields $a + x + z = 180$. Given that $z = 60$ and $a = 20$, it follows that $20 + x + 60 = 180$. Solving for x gives $x = 180 - 60 - 20$, or $x = 100$.

Choice A is incorrect and may result from incorrectly assuming that angles $x + z = 180$. Choice C is incorrect and may result from incorrectly assuming that the smaller triangle is a right triangle, with x as the right angle. Choice D is incorrect and may result from a misunderstanding of the exterior angle theorem and incorrectly assuming that $x = y + z$.